



Yakeen Plus (2025)

PRACTICE TEST - 11

DURATION : 200 Minutes

DATE : 27/11/2024

M. MARKS : 720

Topics Covered

Physics :	Gravitation, Properties of Solids and Liquids, Thermodynamics, Kinetic Theory of Gases, Oscillation and Waves
Chemistry :	Redox Reaction, Electrochemistry, Hydrocarbon, Haloalkanes and Haloarenes, Coordination Compound
Biology :	(Botany) : Morphology of Flowering Plants, Anatomy of Flowering Plants, Respiration in Plants, Photosynthesis in Higher Plants, Plant Growth and Development (Zoology) : Body Fluids and Circulation, Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

SECTION-(I) PHYSICS**SECTION - A**

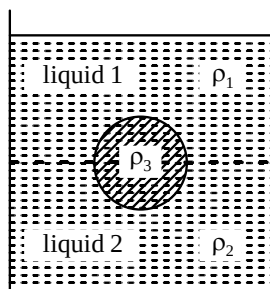
- The distance of the centres of moon and earth is D . The mass of earth is 81 times the mass of the moon. At what distance from the centre of the earth, the gravitational force will be zero.
 - (1) $\frac{D}{2}$
 - (2) $\frac{2D}{3}$
 - (3) $\frac{4D}{3}$
 - (4) $\frac{9D}{10}$
- Mass M is divided into two parts xM and $(1-x)M$. For a given separation, the value of x for which the gravitational attraction between the two pieces becomes maximum is:
 - (1) $\frac{1}{2}$
 - (2) $\frac{3}{5}$
 - (3) 1
 - (4) 2
- If the change in the value of 'g' at a height h above the surface of the earth is the same as at a depth x below it, then (both x and h being much smaller than the radius of the earth):
 - (1) $x = h$
 - (2) $x = 2h$
 - (3) $x = \frac{h}{2}$
 - (4) $x = h^2$
- A body of mass m rises to height $h = R/5$ from the earth's surface, where R is earth's radius. If g is acceleration due to gravity at earth's surface, the increase in potential energy is:
 - (1) mgh
 - (2) $\frac{4}{5}mgh$
 - (3) $\frac{5}{6}mgh$
 - (4) $\frac{6}{7}mgh$
- The gravitational field due to a mass distribution is $E = K/x^3$ in the x -direction. (K is a constant). Taking the gravitational potential to be zero at infinity, its value at a distance x is:
 - (1) K/x
 - (2) $K/2x$
 - (3) K/x^2
 - (4) $K/2x^2$
- Select the **correct** statement from the following:
 - (1) The orbital velocity of a satellite increases with the radius of the orbit.
 - (2) Escape velocity of a particle from the surface of the earth depends on the speed with which it is fired.
 - (3) The time period of a satellite does not depend on the radius of the orbit.
 - (4) The orbital velocity is inversely proportional to the square root of the radius of the orbit.
- For a satellite escape velocity is 11 km/s. If the satellite is launched at an angle of 60° with the vertical, then escape velocity will be:
 - (1) 11 km/s
 - (2) $11\sqrt{3}$ km/s
 - (3) $\frac{11}{\sqrt{3}}$ km/s
 - (4) 33 km/s
- The ratio of the lengths of two wires A and B of same material is 1 : 2 and the ratio of their diameter is 2 : 1. They are stretched by the same force, then the ratio of increase in length will be:
 - (1) 2 : 1
 - (2) 1 : 4
 - (3) 1 : 8
 - (4) 8 : 1
- If a rubber ball is taken at the depth of 200 m in a pool, its volume decreases by 0.1%. If the density of the water is $1 \times 10^3 \text{ kg/m}^3$ and $g = 10 \text{ m/s}^2$, then the volume elasticity in N/m^2 will be:
 - (1) 10^8
 - (2) 2×10^8
 - (3) 10^9
 - (4) 2×10^9
- The stress versus strain graphs for wires of two materials A and B are as shown in the figure. If Y_A and Y_B are the Young's modulus of the materials, then:

 - (1) $Y_B = 2Y_A$
 - (2) $Y_A = Y_B$
 - (3) $Y_B = 3Y_A$
 - (4) $Y_A = 3Y_B$
- The orbital angular momentum of a satellite revolving at a distance r from the centre is L . If the distance is increased to $16r$, then the new angular momentum will be:
 - (1) $16L$
 - (2) $64L$
 - (3) $\frac{L}{4}$
 - (4) $4L$
- A beaker containing a liquid of density ρ moves up with an acceleration ' a '. The pressure due to the liquid at a depth h below free surface of the liquid is:
 - (1) $h\rho g$
 - (2) $h\rho(g-a)$
 - (3) $h\rho(g+a)$
 - (4) $2h\rho g \left(\frac{g-a}{g+a} \right)$

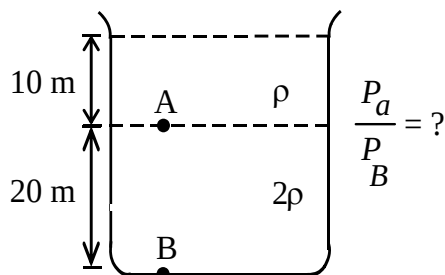
13. Two liquids having densities d_1 and d_2 are mixed in such a way that both have same mass. The density of the mixture is:

(1) $\frac{d_1 + d_2}{2}$ (2) $\frac{d_1 + d_2}{d_1 d_2}$
 (3) $\frac{d_1 d_2}{d_1 + d_2}$ (4) $\frac{2d_1 d_2}{d_1 + d_2}$

14. A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2 respectively. A solid ball, made of a material of density ρ_3 , is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is **true** for ρ_1 , ρ_2 and ρ_3 ?



- (1) $\rho_1 < \rho_3 < \rho_2$ (2) $\rho_3 < \rho_1 < \rho_2$
 (3) $\rho_1 > \rho_3 > \rho_2$ (4) $\rho_1 < \rho_2 < \rho_3$
15. For the given figure ratio of pressure at A and B will be ($\rho = 1000 \text{ kg/m}^3$, $P_0 = 10^5 \text{ Pa}$):

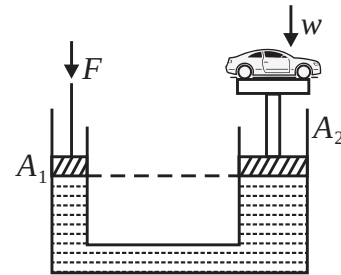


- (1) $\frac{1}{2}$ (2) $\frac{1}{4}$
 (3) $\frac{1}{3}$ (4) $\frac{2}{3}$
16. In the Young's experiment, If length of wire and radius both are doubled then the value of Y will become:
- (1) 2 times (2) 4 times
 (3) Remains same (4) Half

17. Two particles of equal mass go round a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is:

(1) $v = \frac{1}{2R} \sqrt{\frac{1}{Gm}}$ (2) $v = \sqrt{\frac{Gm}{2R}}$
 (3) $v = \frac{1}{2} \sqrt{\frac{Gm}{R}}$ (4) $v = \sqrt{\frac{4Gm}{R}}$

18. In a hydraulic jack as shown, mass of the car $W = 800 \text{ kg}$, $A_1 = 10 \text{ cm}^2$, $A_2 = 10 \text{ m}^2$. The minimum force F required to lift the car is:

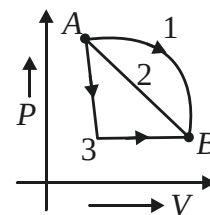


- (1) 1 N (2) 0.8 N
 (3) 8 N (4) 16 N

19. A copper rod length L and radius r is suspended from the ceiling by one of its ends. What will be elongation of the rod due to its own weight when ρ and Y are the density and Young's modulus of the copper respectively?

(1) $\frac{\rho^2 g L^2}{2Y}$ (2) $\frac{\rho g L^2}{2Y}$
 (3) $\frac{\rho^2 g^2 L^2}{2Y}$ (4) $\frac{\rho g L}{2Y}$

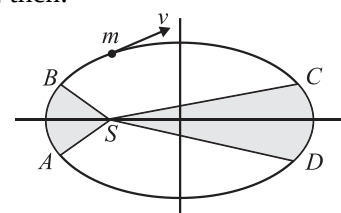
20. An ideal gas goes from state A to state B via three different processes as indicated in the P - V diagram.



If Q_1 , Q_2 , Q_3 indicate the heat absorbed by the gas along the three processes and $\Delta U_1, \Delta U_2, \Delta U_3$ indicate the change in internal energy along the three processes respectively, then

- (1) $Q_1 > Q_2 > Q_3$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$
 (2) $Q_3 > Q_2 > Q_1$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$
 (3) $Q_1 = Q_2 = Q_3$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$
 (4) $Q_3 > Q_2 > Q_1$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$

21. The figure shows elliptical orbit of a planet m about the sun S . The shaded area SCD is twice the shaded area SAB . If t_1 be the time for the planet to move from C to D and t_2 is the time to move from A to B , then:

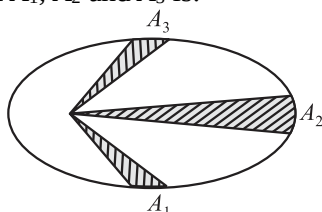


- (1) $t_1 = t_2$ (2) $t_1 = 8t_2$
 (3) $t_1 = 4t_2$ (4) $t_1 = 2t_2$

22. A satellite has kinetic energy K , potential energy U and total energy E . The value of $\frac{K+U+E}{K-U}$ will be:

- (1) $\frac{2}{3}$ (2) 1
(3) $-\frac{2}{3}$ (4) $\frac{3}{2}$

23. A planet moving around the sun sweeps area A_1 in 3 days, A_2 in 6 days and A_3 in 2 days. The relation between A_1 , A_2 and A_3 is:



- (1) $A_1 = A_2 = A_3$ (2) $2A_1 = A_2 = 3A_3$
(3) $4A_1 = A_2 = 6A_3$ (4) $A_1 = 2A_2 = 3A_3$

24. A mass m is at a distance x from one end of a uniform rod of length L and mass M . The gravitational force on the mass due to the rod is:

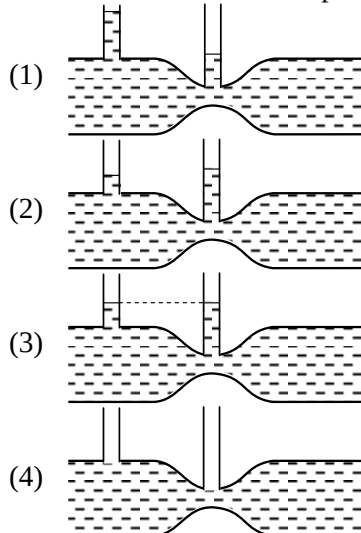


- (1) $\frac{GMm}{(x+L)^2}$ (2) $\frac{GMm}{x(L+x)}$
(3) $\frac{GMm}{x^2}$ (4) $\frac{GMm}{\left(\frac{L}{2} + x\right)^2}$

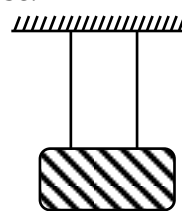
25. If density of earth is ρ , radius is R_e then its gravitational field at distance $R_e/4$ from centre will be:

- (1) $\frac{\pi G \rho R_e}{3}$ (2) $\frac{\pi G \rho R_e}{12}$
(3) $\frac{\pi G \rho R_e}{4}$ (4) $\frac{\pi G \rho R_e}{6}$

26. For a fluid which is flowing steadily, the level in the vertical tubes is best represented by:

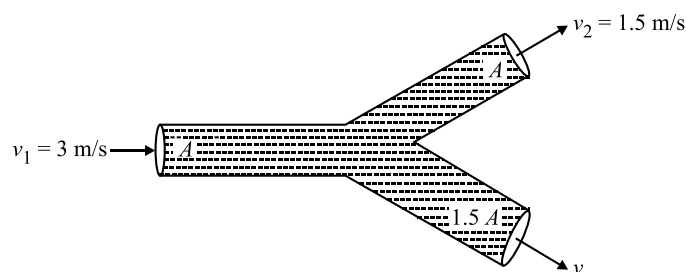


27. Two wires of equal length and cross-section are suspended as shown. Their Young's moduli are Y_1 and Y_2 respectively. The equivalent Young's modulus will be:



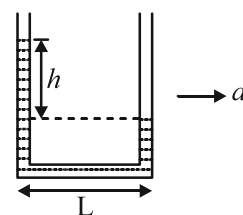
- (1) $Y_1 + Y_2$ (2) $\frac{Y_1 + Y_2}{2}$
(3) $\frac{Y_1 Y_2}{Y_1 + Y_2}$ (4) $\sqrt{Y_1 Y_2}$

28. An incompressible liquid flows through a horizontal tube as shown in the following fig. Then the velocity v of the fluid is:



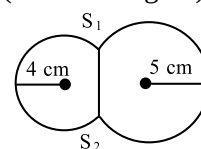
- (1) 3.0 m/s (2) 1.5 m/s
(3) 1.0 m/s (4) 2.25 m/s

29. When at rest, a liquid stands at the same level in the tubes shown in figure. But as indicated a height difference h occurs when the system is given an acceleration a towards the right. Here h is equal to:



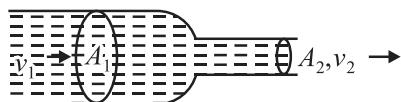
- (1) $\frac{aL}{2g}$ (2) $\frac{gL}{2a}$
(3) $\frac{gL}{a}$ (4) $\frac{aL}{g}$

30. Two soap bubbles of radii r_1 and r_2 equal to 4 cm and 5 cm are touching each other over a common surface S_1S_2 (shown in figure). Its radius will be:



- (1) 4 cm (2) 20 cm
(3) 5 cm (4) 4.5 cm

31. A liquid flows in the tube from left to right as shown in figure. A_1 and A_2 are the cross-section of the portion of the tube as shown. The ratio of speed v_1 / v_2 will be:



- (1) $\frac{A_1}{A_2}$
- (2) $\frac{A_2}{A_1}$
- (3) $\sqrt{\frac{A_2}{A_1}}$
- (4) $\sqrt{\frac{A_1}{A_2}}$

32. In thermodynamic processes which of the following statements is **not** true?

- (1) In an isochoric process pressure remains constant.
- (2) In an isothermal process the temperature remains constant.
- (3) In an adiabatic process $PV^\gamma = \text{constant}$.
- (4) In an adiabatic process the system is insulated from the surroundings.

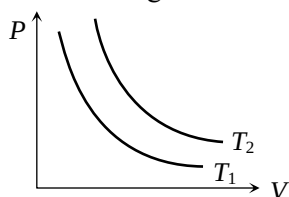
33. The S.H.M. of a particle is given by the equation $y = 3\sin \omega t + 4\cos \omega t$. The amplitude is:

- (1) 7
- (2) 1
- (3) 5
- (4) 12

34. In thermodynamic process, 200 Joules of heat is given to a gas and 100 Joules of work is also done on it. The change in internal energy of the gas is:

- (1) 100 J
- (2) 300 J
- (3) 419 J
- (4) 24 J

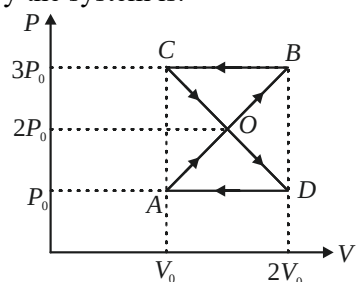
35. The adjoining figure shows graph of pressure and volume of a gas at two temperatures T_1 and T_2 . Which of the following interferences is **correct**:



- (1) $T_1 > T_2$
- (2) $T_1 = T_2$
- (3) $T_1 < T_2$
- (4) No interference can be drawn

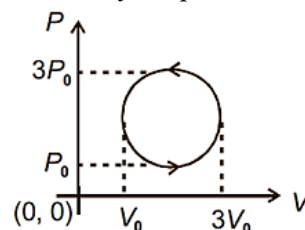
SECTION - B

36. A thermodynamic system undergoes cyclic process $ABCD$ as shown in figure. The work done by the system is:



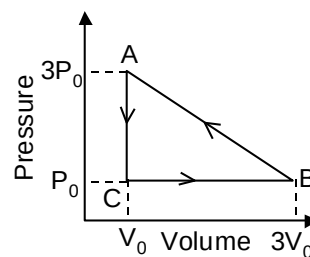
- (1) P_0V_0
- (2) $2P_0V_0$
- (3) $\frac{P_0V_0}{2}$
- (4) Zero

37. Work done in the cyclic process shown in figure is:



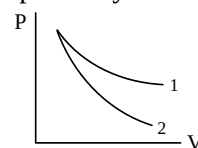
- (1) $4P_0V_0$
- (2) $-4P_0V_0$
- (3) $-\frac{22}{7}P_0V_0$
- (4) $-13P_0V_0$

38. In the figure shown here, the work done in the process $ACBA$ is:



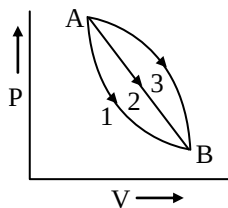
- (1) $4P_0V_0$
- (2) $6P_0V_0$
- (3) $-2P_0V_0$
- (4) $-4P_0V_0$

39. P-V plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond respectively to:

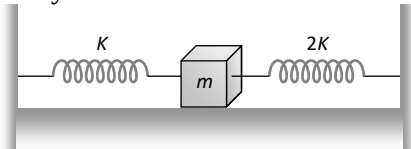


- (1) He and O_2
- (2) O_2 and He
- (3) He and Ar
- (4) O_2 and N_2

40. A gas of given mass, is brought from stage A to B along three paths 1, 2 and 3, as shown in the figure. If the amount of work done in these three processes is respectively equal to W_1 , W_2 and W_3 , then:



- (1) $W_1 > W_2 > W_3$
 - (2) $W_1 < W_2 < W_3$
 - (3) $W_1 = W_2 = W_3$
 - (4) $W_1 < W_2$, $W_3 < W_2$
41. Two springs of force constants K and $2K$ are connected to a mass as shown below. The frequency of oscillation of the mass is



- (1) $(1/2\pi)\sqrt{(K/m)}$
 - (2) $(1/2\pi)\sqrt{(2K/m)}$
 - (3) $(1/2\pi)\sqrt{(3K/m)}$
 - (4) $(1/2\pi)\sqrt{(m/K)}$
42. Which of the following statements about kinetic theory of gases is **wrong**?
- (1) The molecules of a gas are in continuous random motion.
 - (2) The molecules continuously undergo inelastic collisions.
 - (3) The molecules do not interact with each other except during collisions.
 - (4) The collisions amongst the molecules are of short duration.
43. At a certain temperature, the r.m.s. velocity for O_2 is 400 m/sec. At the same temperature, the r.m.s. velocity for H_2 molecules will be:
- (1) 100 m/sec
 - (2) 25 m/sec
 - (3) 1600 m/sec
 - (4) 6400 m/sec

44. A transverse wave propagating along x -axis is represented by: $y(x, t) = 8 \sin(0.5\pi x - 4\pi t - \frac{\pi}{4})$ where x is in metres and t is in seconds. The speed of the wave is:
- (1) 4π m/s
 - (2) 0.5π m/s
 - (3) $\frac{\pi}{4}$ m/s
 - (4) 8 m/s

45. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two.

The lowest resonant frequency for this string is:

- (1) 105 Hz
- (2) 155 Hz
- (3) 205 Hz
- (4) 10.5 Hz

46. If n_1 , n_2 and n_3 are the fundamental frequencies of three segments into which a string is divided, then the original fundamental frequency n of the string is given by:

- (1) $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$
- (2) $\frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n_1}} + \frac{1}{\sqrt{n_2}} + \frac{1}{\sqrt{n_3}}$
- (3) $\sqrt{n} = \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3}$
- (4) $n = n_1 + n_2 + n_3$

47. The equation of a simple harmonic wave is given by $y = 3\sin\frac{\pi}{2}(50t - x)$, where x and y are in metres and t is in seconds. The ratio of maximum particle velocity of the wave velocity is:

- (1) 2π
- (2) $\frac{3}{2}\pi$
- (3) 3π
- (4) $2/3\pi$

48. A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm of column length.

If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is:

- (1) 330 m/s
- (2) 339 m/s
- (3) 300 m/s
- (4) 350 m/s

49. The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system?

- (1) 10 Hz
- (2) 20 Hz
- (3) 30 Hz
- (4) 40 Hz

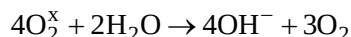
50. Equation of a transverse wave pulse is $y = \frac{9}{3 + (x - 10t)^2}$, (where x and y are in metre and t is in second). The speed of the wave pulse is:

- (1) 3 m/s
- (2) 5 m/s
- (3) 10 m/s
- (4) 100 m/s

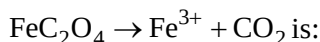
SECTION-(II) CHEMISTRY

SECTION - A

51. In the balanced reaction



- (1) $x = -4$ and species is oxide
- (2) $x = -2$ and species is superoxide
- (3) $x = 0$ and species is oxygen
- (4) $x = -1$ and species is superoxide

52. Equivalent weight of FeC_2O_4 in the change:(Where molecular mass of FeC_2O_4 is M)

- (1) $M/3$
- (2) $M/6$
- (3) $M/2$
- (4) $M/1$

53. What is the oxidation state of S in Na_2S_2 ?

- (1) +1
- (2) -2
- (3) -1
- (4) 0

54. The reaction, $\text{H}_2\text{S} + \text{H}_2\text{O}_2 \rightarrow \text{S} + 2\text{H}_2\text{O}$ manifests:

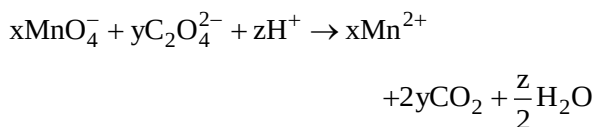
- (1) oxidising action of H_2O_2
- (2) reducing nature of H_2O_2
- (3) acidic nature of H_2O_2
- (4) alkaline nature of H_2O_2

55. Which of the following is **correct**?

(A)	Oxidation	=	addition of oxygen
(B)	Oxidation	=	addition of electronegative element
(C)	Oxidation	=	removal of hydrogen
(D)	Oxidation	=	removal of electronegative element

- (1) A, B, C
- (2) B, C, D
- (3) A, C, D
- (4) A, B, C, D

56. Consider the following reaction:



The value of x, y and z in the reaction are respectively

- (1) 5, 2 and 6
- (2) 2, 5 and 8
- (3) 2, 5 and 16
- (4) 5, 2 and 8

57. Match List-I with List-II.

List-I (Type of reaction)		List-II (Example)	
(A)	Combination reactions	(I)	$\text{P}_4(\text{s}) + 3\text{OH}^-(\text{aq}) + 3\text{H}_2\text{O} \rightarrow \text{PH}_3(\text{g}) + 3\text{H}_2\text{PO}_2^-(\text{aq})$
(B)	Decomposition reactions	(II)	$\text{C}(\text{s}) + \text{O}_2(\text{g}) \xrightarrow{\Delta} \text{CO}_2(\text{g})$
(C)	Non-metal displacement reactions	(III)	$2\text{NaH}(\text{s}) \xrightarrow{\Delta} 2\text{Na}(\text{s}) + \text{H}_2(\text{g})$
(D)	Disproportionation reactions	(IV)	$2\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq}) + \text{H}_2(\text{g})$

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-I, B-II, C-IV, D-III
- (3) A-II, B-I, C-III, D-IV
- (4) A-I, B-II, C-III, D-IV

58. According to crystal field theory, the M-L interaction in a complex is:

- (1) Purely ionic
- (2) Purely coordinate
- (3) Purely covalent
- (4) Partially covalent

59. Molar conductance of a complex of cobalt is zero. Then its structure would be:

- (1) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- (2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
- (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
- (4) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$

60. Which of the following is **incorrect** for $[\text{CoF}_6]^{3-}$?

- (1) It has two unpaired electrons.
- (2) It is paramagnetic.
- (3) Co in $[\text{CoF}_6]^{3-}$ is sp^3d^2 hybridised.
- (4) It has octahedral geometry.

61. When two half-cells of electrode potential of E_1 and E_2 are combined to form a cell of electrode potential E_3 , then (Where n_1 , n_2 and n_3 are no. of electrons exchanged in first, second and combined half-cells) give an expression for E_3 using above informations:

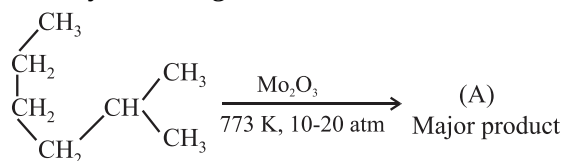
- (1) $E_3 = E_2 - E_1$
- (2) $E_3 = \frac{E_1 n_1 + E_2 n_2}{n_3}$
- (3) $E_3 = E_1 n_1 - E_2 n_2$
- (4) $E_3 = E_1 + E_2$

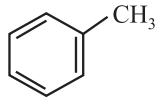
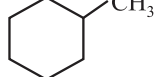
62. During electrolysis of an aqueous solution of sodium sulphate if 2.4 L of oxygen at STP was liberated at anode. The volume of hydrogen at STP, liberated at cathode would be:
 (1) 1.2 L
 (2) 2.4 L
 (3) 2.6 L
 (4) 4.8 L
63. Λ_m° value cannot be determined by extrapolating, the plot between Λ_m verses $C^{1/2}$ for the compound
 (1) KCl
 (2) K_2SO_4
 (3) NH_4OH
 (4) $NaNO_3$
64. The standard reduction potentials of $Cu^{2+} | Cu$ and $Cu^{2+} | Cu^+$ are 0.337 and 0.153 V respectively. The standard electrode potential of $Cu^+ | Cu$ half cell is:
 (1) 0.184 V
 (2) 0.827 V
 (3) 0.521 V
 (4) 0.490 V
65. Voltage of the cell where distance between the electrodes is unity is known as
 (1) Cell emf
 (2) Equivalent conductivity
 (3) Potential gradient
 (4) Ionic mobility
66. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R
Assertion A: Molar conductivity increases with decrease in concentration.
Reason R: On decreasing the concentration, the total volume (V), of solution containing one mole of electrolyte also decreases.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
67. The tetrahedral crystal field splitting is only _____ of the octahedral splitting.
 (1) 1/9
 (2) 2/9
 (3) 4/9
 (4) 5/9
68. Which of the following is a hexadentate ligand?
 (1) NH_3
 (2) Cl^-
 (3) EDTA
 (4) dien
69. Which of the following does **not** belongs to double salt?
 (1) $K_4[Fe(CN)_6]$
 (2) $KCl \cdot MgCl_2 \cdot 6H_2O$
 (3) $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$
 (4) $KAl(SO_4)_2 \cdot 12H_2O$
70. In $[Ni(NH_3)_4]SO_4$, the primary valency and coordination number of Ni will be respectively
 (1) 3 and 6
 (2) 4 and 4
 (3) 4 and 2
 (4) 2 and 4
71. When an alkene on ozonolysis gives formaldehyde and acetaldehyde, the alkene is:
 (1) Ethene
 (2) Propene
 (3) But-1-ene
 (4) But-2-ene
72. The major product formed in the given reaction is:

$$CH_3 - CH_2 - \underset{\substack{| \\ OH}}{CH} - CH_3 \xrightarrow[\Delta]{Conc. H_2SO_4}$$

 (1) $CH_3 - CH_2 - CH = CH_2$
 (2) $CH_3 - CH = CH - CH_3$
 (3) $CH_3 - \underset{\substack{\diagdown \\ O \\ \diagup}}{CH} - CH - CH_3$
 (4) $CH_3 - CH_2 - \underset{\substack{|| \\ O}}{C} - CH_3$
73. When 2-butyne reacts with sodamide in an inert solvent in the presence of dilute HCl, the product formed is:
 (1) n-Butane
 (2) 2-Butene
 (3) 1-Butyne
 (4) 1-Propyne

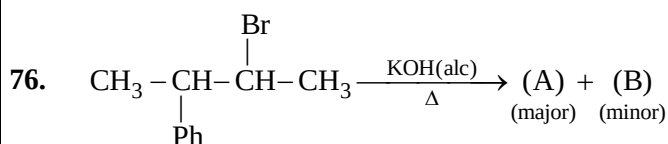
74. Identify A in the given chemical reaction.



- (1)  (2) 
 (3)  (4) 

75. A gaseous hydrocarbon X on reaction with bromine in light forms a mixture of two monobromo alkanes and HBr. The hydrocarbon X is:

- (1) C_6H_6 (2) C_3H_6
 (3) C_3H_8 (4) C_5H_{12}



Major product will be:

- (1) $\text{CH}_3 - \text{C} = \text{CH} - \text{CH}_3$
 |
 Ph
 (2) $\text{CH}_3 - \text{CH} - \text{CH} = \text{CH}_2$
 |
 Ph
 (3) $\text{CH}_2 = \text{C} - \text{CH}_2 - \text{CH}_3$
 |
 Ph
 (4) $\text{CH}_2 = \text{C} - \text{CH}_2 - \text{Ph}$
 |
 Ph

77. Given below are two statements:

Statement I: Tertiary alkyl halides undergo faster $\text{S}_{\text{N}}1$ reactions than primary alkyl halides.

Statement II: The stability of the carbocation intermediate is a major factor in the rate of $\text{S}_{\text{N}}1$ reactions.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

78. 2, 2- dichlorobutane on boiling with aqueous KOH gives

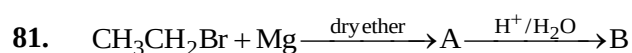
- (1) Butanal
 (2) 2-Butanone
 (3) 2- Butanol
 (4) Butanoic acid

79. Cyclopentene on treatment with alkaline KMnO_4 gives

- (1) cis-cyclopentane-1,2-diol
 (2) trans-cyclopentane-1,2-diol
 (3) Cyclopentanol
 (4) 1:1 mixture of cis and trans 1, 2 - cyclopentanediol

80. Which of the following has minimum boiling point?

- (1) Isobutene
 (2) n-Butane
 (3) 1-Butyne
 (4) 1-Butene



The product B in the above reaction sequence is:

- (1) CH_3CH_3
 (2) $\text{CH}_2 = \text{CH}_2$
 (3) $\text{CH} \equiv \text{CH}$
 (4) $\text{CH}_3\text{CH}_2\text{OH}$

82. IUPAC name of $(\text{CH}_3)_3\text{CCl}$ is:

- (1) 3-Chlorobutane
 (2) 2-Chloro-2-methylpropane
 (3) t-Butyl chloride
 (4) n-Butyl chloride

83. The Friedel-Crafts reaction of n-propyl bromide with benzene in the presence of anhydrous AlCl_3 gives:

- (1) n-Propyl benzene
 (2) 1,4-Dipropyl benzene
 (3) Isopropyl benzene
 (4) 1,2 -Dipropyl benzene

84. CHCl_3 (Chloroform) is stored in closed dark coloured bottles and completely filled.....
- (1) As it is decomposed to give CH_2Cl_2 and Cl_2 in presence of light and Cl_2 is poisonous.
 - (2) As it is decomposed to give $(\text{CCl}_4 \text{ and } \text{HCl})$ in presence of light and HCl is poisonous and acidic as well.
 - (3) As it slowly oxidized by air in presence of light to an extremely poisonous gas, COCl_2 (Phosgene).
 - (4) As it is slowly oxidized to give CO and HCl , which are poisonous.
85. Fluorobenzene ($\text{C}_6\text{H}_5\text{F}$) can be synthesised in the laboratory:
- (1) By heating phenol with HF and KF .
 - (2) From aniline by diazotisation followed by heating the diazonium salt with HBF_4 .
 - (3) By direct fluorination of benzene with F_2 gas.
 - (4) By reacting bromobenzene with NaF solution.

SECTION - B

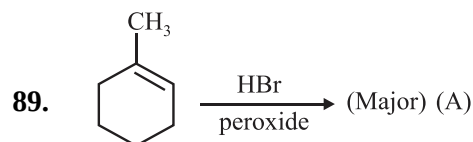
86. Given below are two statements:
- Statement I:** Boiling points of isomeric dichlorobenzene are very nearly the same.
- Statement II:** The boiling points of isomeric haloalkanes increases with increase in branching.
- In the light of above statements, choose the **correct** answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
87. With respect to the starting material, $\text{CH}_2 = \text{CH} - \text{CH}_3$, match the **List-I** with **List-II**

List-I (Reagents)		List-II (Reaction)	
(A)	$\text{KMnO}_4/\text{H}_2\text{O}$	(I)	Anti-Markovnikov addition
(B)	Br_2	(II)	Oxidative cleavage
(C)	$\text{HBr}/\text{R}_2\text{O}_2$	(III)	Syn addition
(D)	Hot KMnO_4/H^+	(IV)	Anti-addition

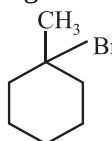
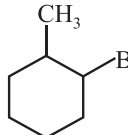
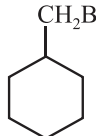
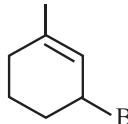
Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-I, B-II, C-III, D-IV

88. Isopropyl bromide on Wurtz reaction gives
- (1) Hexane
 - (2) Propane
 - (3) 2, 3-Dimethylbutane
 - (4) Neo-hexane



In the given reaction, identify A.

- (1) 
- (2) 
- (3) 
- (4) 

90. Which of the following alkyl halide is iso-butyl bromide?

- (1) $\text{CH}_3\text{CH}_2\underset{\text{Br}}{\text{CH}}\text{CH}_3$
- (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
- (3) $\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{Br}$
- (4) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2\text{Br}$

91. Which of the following arrangements represents **correct** order of increasing oxidation number of the central atom?

- (1) $\text{CO}_3^{2-} < \text{ClO}_3^- < \text{CrO}_4^{2-} < \text{MnO}_4^-$
- (2) $\text{ClO}_3^- < \text{CrO}_4^{2-} < \text{MnO}_4^- < \text{CO}_3^{2-}$
- (3) $\text{CO}_3^{2-} < \text{ClO}_3^- < \text{MnO}_4^- < \text{CrO}_4^{2-}$
- (4) $\text{CrO}_4^{2-} < \text{MnO}_4^- < \text{CO}_3^{2-} < \text{ClO}_3^-$

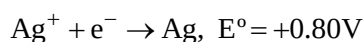
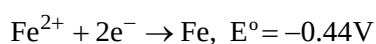
92. Oxidation state of carbon in HCOOH will be:

- (1) +1
- (2) +2
- (3) -4
- (4) 0

93. Which of the following is a redox reaction?

- (1) $\text{NaCl} + \text{KNO}_3 \rightarrow \text{NaNO}_3 + \text{KCl}$
- (2) $\text{CaC}_2\text{O}_4 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{C}_2\text{O}_4$
- (3) $\text{Mg}(\text{OH})_2 + 2\text{NH}_4\text{Cl} \rightarrow \text{MgCl}_2 + 2\text{NH}_4\text{OH}$
- (4) $\text{Zn} + 2\text{AgCN} \rightarrow 2\text{Ag} + \text{Zn}(\text{CN})_2$

94. Using the standard potential values given below, decide which of the statements A, B, C, D are **correct**:



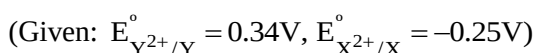
- A. Copper can displace iron from FeSO_4 solution.
- B. Iron can displace copper from CuSO_4 solution.
- C. Silver can displace Cu from CuSO_4 solution.
- D. Iron can displace silver from AgNO_3 solution.

- (1) A and B
- (2) B and C
- (3) B and D
- (4) A and D

95. An electrochemical cell stops working after some times because:

- (1) Electrode potential of both the electrodes becomes zero.
- (2) Electrode potential of both electrodes becomes equal.
- (3) One of the electrode is eaten away.
- (4) The reaction starts proceeding in opposite directions.

96. If a half-cell X/X^{2+} (0.1 M) is connected to another half-cell Y/Y^{2+} (1.0 M) By means of a salt bridge and an external circuit at 25°C , the cell voltage would be:



- (1) 0.26 V
- (2) 0.31 V
- (3) 0.62 V
- (4) 0.72 V

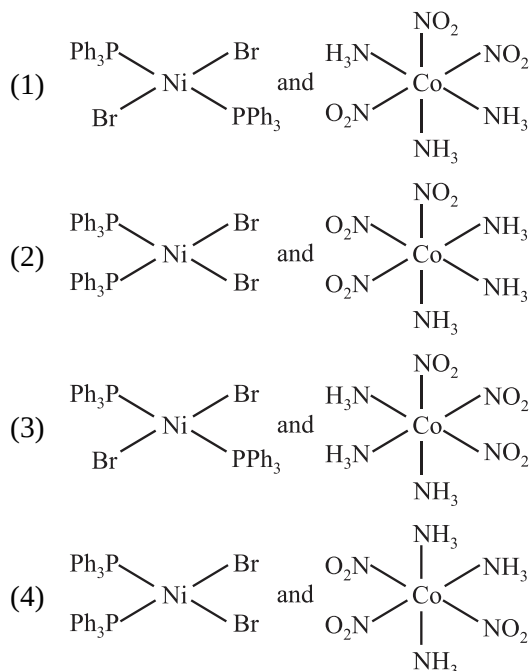
97. During electrolysis of fused sodium chloride:

- (1) Chloride ions are oxidised at cathode
- (2) Chloride ions are oxidised at anode
- (3) Sodium ions are oxidised at cathode
- (4) Sodium ions are oxidised at anode

98. If $\Delta_0 < P$ (pairing energy), the **correct** electron configuration for d^4 system in octahedral crystal will be:

- (1) t_{2g}^4, e_g^0
- (2) t_{2g}^3, e_g^1
- (3) t_{2g}^0, e_g^4
- (4) t_{2g}^0, e_g^2

99. The **correct** structures of trans- $[\text{NiBr}_2(\text{PPh}_3)_2]$ and meridional $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$, respectively, are:



100. Which of the following complexes is used to be as an anticancer agent?

- (1) Cis- $[\text{PtCl}_2(\text{NH}_3)_2]$
- (2) Cis- $\text{K}_2[\text{PtCl}_2\text{Br}_2]$
- (3) $\text{Na}_2[\text{CoCl}_4]$
- (4) Mer- $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$

SECTION-(III) BOTANY**SECTION - A**

- 101.** The cells of xylem parenchyma:
- (1) are dead and thin walled.
 - (2) are living and thick walled.
 - (3) do not store food in any form.
 - (4) are living and thin walled.
- 102.** Which of the below feature(s) belong(s) to monocot leaves?
- A. Stomata are present on both the surfaces of the epidermis.
 - B. Presence of cuticle.
 - C. Bulliform cells.
 - D. Mesophyll is differentiated into palisade and spongy parenchyma.
- Choose the most appropriate answer from the options given below:
- (1) D only
 - (2) A and D only
 - (3) A, B and C only
 - (4) A, B, C and D
- 103.** Palisade parenchyma is **present**:
- (1) below the abaxial epidermis in monocot leaves.
 - (2) below the adaxial epidermis in monocot leaves.
 - (3) around the vascular bundles in dicot leaves and monocot leaves.
 - (4) below the adaxial epidermis in dicot leaves.
- 104.** Monocot root differs from dicot root in having:
- (1) endodermis that gives rise to lateral roots.
 - (2) more than 6 vascular bundles.
 - (3) small and inconspicuous pith.
 - (4) casparian strips in the endodermis.
- 105.** Among the following anatomical features:
- A. enclose stomatal pore.
 - B. are dumb bell shaped in dicots.
 - C. do not have chloroplast.
 - D. regulate opening and closing of stomata.
- How many are **correct** for guard cells?
- (1) Four
 - (2) Three
 - (3) Two
 - (4) One
- 106.** Pericycle forms semi lunar patches of sclerenchyma cells in:
- (1) dicot roots.
 - (2) monocots stems.
 - (3) dicot stems.
 - (4) monocot roots.

107. Match List-I with List-II:

	List-I		List-II
(A)	Dicot stem	(I)	Pith is small
(B)	Monocot root	(II)	Mesophyll
(C)	Leaves	(III)	Hypodermis present
(D)	Dicot root	(IV)	Pith is well developed

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-IV, B-II, C-I, D-III
- (4) A-III, B-IV, C-I, D-II

108. Read the following statement and select the correct option.

- A. From the region of elongation, some of the epidermal cells form root hairs.
- B. Adventitious roots are seen in banyan trees.
- C. Maize and sugarcane have prop roots.
- D. The root is covered at the apex by a thimble-like structure called the root cap.

Choose the most appropriate answer from the options given below:

- (1) B only
- (2) B and D only
- (3) A, B and C only
- (4) A, B, C and D

109. Match List-I with List-II:

	List-I		List-II
(A)	<i>Calotropis</i>	(I)	Alternate phyllotaxy
(B)	Sunflower	(II)	Pinnately compound leaf
(C)	<i>Alstonia</i>	(III)	Opposite phyllotaxy
(D)	Rachis	(IV)	Whorled phyllotaxy

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-III, B-IV, C-I, D-II
- (3) A-IV, B-II, C-I, D-III
- (4) A-II, B-IV, C-III, D-I

110. Given figure of root is of:



- (1) adventitious roots of maize.
(2) tap root of wheat.
(3) fibrous roots of wheat.
(4) adventitious roots of grasses.
111. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Plants growing in swampy areas possess pneumatophores.
Reason R: Pneumatophores help the plant to get oxygen for respiration.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.
112. Given below are two statements:
Statement I: Androecium and gynoecium are reproductive organs of flower.
Statement II: When a flower either has only stamens or only carpels it is said to be unisexual.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.
113. Read the following statements:
A. A flower may be trimerous, tetramerous or pentamerous when the floral appendages are in multiples of 3, 4 or 5.
B. Flowers with bracts-reduced leaf found at the base of the pedicel.
C. The stalk of flower is called pedicel.
D. Petals are usually brightly coloured to attract insects for pollination.
How many of the above statement(s) is/are **correct**?
(1) Three (2) Two
(3) One (4) Four

114. Identify the family with the following characters:
A. Leaves having alternate phyllotaxy.
B. Bisexual and actinomorphic flowers.
C. Five epipetalous stamens.
D. Fruits are berry or capsule.
E. Seeds are endospermic.
(1) Solanaceae
(2) Liliaceae
(3) Fabaceae
(4) Brassicaceae
115. In glycolysis, during conversion of 2-phosphoglycerate to phosphoenolpyruvate:
(1) a molecule of ATP is synthesised.
(2) a water molecule is released.
(3) decarboxylation reaction occurs.
(4) a molecule of $\text{NADH} + \text{H}^+$ is released.
116. How does alcohol fermentation differ from lactic acid fermentation?
(1) Is hazardous.
(2) Produces 2 ATP as net gain.
(3) Is involved with reoxidation of $\text{NADH} + \text{H}^+$ to NAD^+ .
(4) Releases CO_2 along with ethanol.
117. Given below are two statements:
Statement I: Krebs' cycle takes place in the matrix of mitochondria.
Statement II: Pyruvate is transported from the cytoplasm into the mitochondria.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.
118. Which of the following molecules directly enter(s) the glycolytic pathway?
A. Sucrose
B. Glucose
C. Cellulose
D. Fructose
Choose the most appropriate answer from the options given below:
(1) B only
(2) A, B and D only
(3) B and D only
(4) A and C only

119. _____ is the small protein to outer surface of inner mitochondrial membrane and acts as mobile carrier for transfer of electrons between complex III and IV.

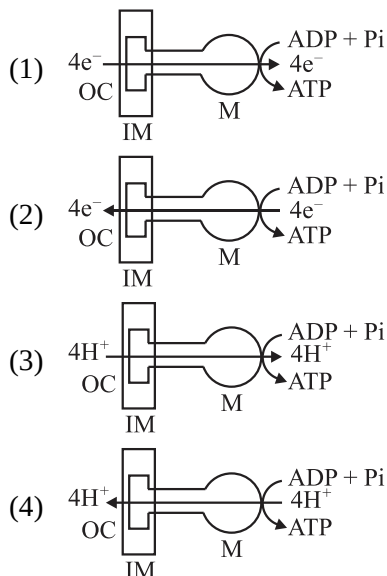
- (1) Cytochrome *a* (2) Cytochrome *c*
(3) Cytochrome *a*₃ (4) Ubiquinone

120. Which of the following diagram represents the synthesis of ATP in mitochondria?

I.M = Inner membrane

M = Matrix

O.C = Outer Chamber



121. What is **true** about glycolysis?

- (1) In this process, glucose undergoes partial oxidation to form two molecules of pyruvic acid.
(2) In this process, glucose undergoes complete oxidation to form two molecules of pyruvic acid.
(3) In this process, glucose undergoes partial oxidation to form two molecules of carbon dioxide.
(4) In this process, glucose undergoes complete oxidation to form two molecules of carbon dioxide.

122. How many ATP are directly synthesised in glycolysis from one glucose molecule?

- (1) Two (2) Three
(3) One (4) Four

123. The foolish seedling disease in rice plant:

- (1) occurred due to high concentration of auxin.
(2) occurred due to high concentration of ABA.
(3) caused by a fungus *Gibberella fujikuroi*.
(4) induced by a hormone cytokinin.

124. Match **List-I** with **List-II**:

	List-I		List-II
(A)	Lag phase	(I)	Growth is rapid and maximum
(B)	Exponential phase	(II)	Growth slows down and become steady
(C)	Stationary phase	(III)	Initial phase of growth
(D)	Relative growth	(IV)	Efficiency index

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-II, B-III, C-I, D-IV

125. Select the **incorrect** match.

- (1) Skoog and Miller – Identified a substance that is adenine derivatives.
(2) Cousins – Confirmed the release of volatile substance that hastens the ripening of fruits.
(3) F.W. Went – He observed symptoms of disease caused by *Gibberella fujikuroi* in rice seedlings.
(4) Charles Darwin – Observed response of canary grass coleoptile to unilateral illumination of light.

126. Abscission of leaves, fruits and flowers is promoted by:

- A. cytokinin. B. ethylene.
C. gibberellins. D. abscisic acid.

Choose the most appropriate answer from the options given below:

- (1) D only
(2) B and C only
(3) A, B and C only
(4) B and D only

127. Match **List-I** with **List-II**:

	List-I		List-II
(A)	Auxin	(I)	Increases size of apple
(B)	Absciscic acid	(II)	Control xylem differentiation
(C)	Gibberellin	(III)	Closure of stomata
(D)	Ethylene	(IV)	Fruit ripening

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
(2) A-III, B-I, C-IV, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-III, C-II, D-I

- 128.** ABA helps seeds to withstand desiccation by:
- (1) acting as antagonist to cytokinin.
 - (2) stimulating stomatal closure.
 - (3) inducing seed dormancy.
 - (4) playing important role in seed development.
- 129.** The plant growth hormone which is a purine derivative helps?
- (1) To elongate and improve the shape of fruits.
 - (2) To prepare weed free lawns.
 - (3) In adventitious shoot formation.
 - (4) In promoting flowering.
- 130.** If the apical bud is removed from a plant then what will happen to it?
- (1) It will show yellowing of leaves.
 - (2) Its growth will stop.
 - (3) It will stop flowering and becomes deficient in zinc.
 - (4) It will show more lateral shoot growth.
- 131.** Read the statements carefully and choose the **correct** option.
- A. C_3 plant is well adapted to tolerate high temperature.
 - B. Bundle sheath cells of maize leaves are larger, thick walled, and without intercellular spaces.
 - C. Breakdown of C_4 -acid of C_4 -pathway occurs in bundle sheath cells.
 - D. C_4 -plants shows double fixation of CO_2 .
- Choose the most appropriate answer from the options given below:
- (1) Only A, B and D are correct
 - (2) Only B, C and D are correct
 - (3) Only A and D are incorrect
 - (4) Only B and C are incorrect

- 132.** In photosynthetic pathway of sorghum plant C_4 -acids are decarboxylated in **A** to start enzyme activity is **B**. Choose the **correct** options that fills **A** and **B**.

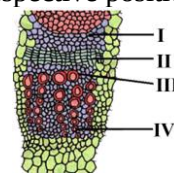
A	B
(1) mesophyll	PEPcase
(2) bundle sheath cell	RuBisCO
(3) mesophyll	RuBisCO
(4) bundle sheath cell	PEPcase

- 133.** To fix 24 molecules of carbon dioxide into sugar in tomato plant, the number of NADPH and ATP required are respectively.
- (1) 48 and 48
 - (2) 48 and 72
 - (3) 48 and 24
 - (4) 72 and 48

- 134.** Select the **incorrect** one w.r.t ATP synthase enzyme present in chloroplast.
- (1) CF_1 head has catalytic activity.
 - (2) CF_0 protrude on the outer surface of thylakoid membrane.
 - (3) The proton gradient is broken down due to the movement of protons across ATP synthase.
 - (4) CF_0 has the proton channel for proton movement.
- 135.** During ETS of photosynthesis, H^+ ions are:
- (1) pumped out from thylakoid lumen always.
 - (2) high in concentration inside the thylakoid lumen.
 - (3) not required to form ATP.
 - (4) always in high concentration in stroma.

SECTION - B

- 136.** Bulliform cells are concerned with:
- (1) conversion of glucose into starch.
 - (2) rolling of leaves during water stress.
 - (3) storage of food material.
 - (4) synthesis of resins and tannins.
- 137.** Read the following statements:
- A. Conjoint type of vascular bundles is common in stems and leaves.
 - B. The monocots and dicots are seen to be anatomically different.
 - C. Epidermis is usually multilayered.
 - D. The cells of epidermis bear a number of hairs.
- How many is/are **incorrect**?
- (1) One
 - (2) Two
 - (3) Three
 - (4) Four
- 138.** Exarch and endarch condition is defined on the basis of the respective position of:



- (1) I and II
- (2) II and III
- (3) III and IV
- (4) IV and I

- 139.** In roots of flowering plants:
- (1) the protoxylem lies towards the pith.
 - (2) arrangement of primary xylem is called exarch type.
 - (3) the metaxylem lies towards the periphery of the organ.
 - (4) only metaxylem is present and no protoxylem is found.
- 140.** Which of the given function is **not** under influence of gibberellins?
- (1) Bolting in rosette plants.
 - (2) Induction of root hair development on stem cutting.
 - (3) Promotes flowering in pineapples.
 - (4) Both (2) and (3)

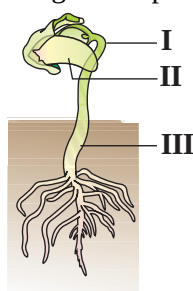
141. Select the **correct** statement.

- (1) NAA is a synthetic auxin.
- (2) Auxins promote fruit and leaf drop at an early stage.
- (3) Auxins prevent the abscission of older mature leaves and fruits.
- (4) 2,4-D is a natural auxin.

142. Which of the following phytohormones is both a growth promoter and growth inhibitor?

- (1) Ethylene
- (2) Gibberellin
- (3) Cytokinin
- (4) ABA

143. Choose **correct** set of labels for given diagram showing a seedling development.



	I	II	III
(1)	Epicotyl	Hypocotyl	Radicle
(2)	Hypocotyl	Hypocotyl	Radicle
(3)	Epicotyl hook	Cotyledons	Hypocotyl
(4)	Epicotyl hook	Cotyledons	Epicotyl

144. Given below are two statements:

Statement I: There can be a net gain of 38 ATP molecules during aerobic respiration of one molecule of glucose.

Statement II: Pure proteins or fats favoured substrate for respiration.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

145. How many molecules of CO_2 are evolved when two molecules of tripalmitin are used as respiratory substrate to measure its RQ value?

- (1) 145 molecules
- (2) 45 molecules
- (3) 102 molecules
- (4) 6 molecules

146. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Chlorophyll *b*, xanthophylls and carotenoids are called as accessory pigments.

Reason R: They absorb light and transfer the energy to chlorophyll *a* and protect from photo-oxidation.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

147. RuBisCO enzyme is:

- (1) found only in mesophyll cell of C_4 plants.
- (2) present in the both C_3 and C_4 plants.
- (3) having greater affinity for oxygen when CO_2 concentration is more than O_2 .
- (4) the least abundant enzyme on earth.

148. Match **List-I** with **List-II**:

List-I (Aestivation)		List-II (Example)	
(A)	Imbricate	(I)	<i>Calotropis</i>
(B)	Valvate	(II)	Gulmohur
(C)	Vexillary	(III)	Cotton
(D)	Twisted	(IV)	Bean

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-IV, C-III, D-I
- (4) A-I, B-III, C-IV, D-II

149. Select the **correct** statement regarding the Malvaceae family.

- (1) Dicotyledonous seeds, non-endospermic, and leaves with parallel venation.
- (2) Flowers are epipetalous, and the ovary is bicarpellary and syncarpous.
- (3) Polypetalous flowers with twisted aestivation.
- (4) Flowers are bracteate, zygomorphic and bisexual.

150. A flower is a modified shoot wherein the shoot apical meristem changes to X . Y do not elongate and the axis gets condensed. Choose the **correct** options that fills X and Y.

X	Y
(1) floral meristem	internodes
(2) intercalary meristem	internodes
(3) floral meristem	nodes
(4) intercalary meristem	nodes

SECTION-(IV) ZOOLOGY**SECTION - A**

151. Choose the **correct** statement w.r.t excretory system.

- (1) Skin and ovaries assist in excretion.
- (2) Nearly 9.1% of filtrate is reabsorbed through different parts of nephron.
- (3) Kidneys are situated close to the dorsal inner wall of the abdominal cavity.
- (4) Nitrogenous wastes like urea are added to dialysing fluid to maintain a concentration gradient.

152. The part of renal tubule which is almost impermeable to electrolytes is:

- (1) descending limb of loop of Henle.
- (2) ascending limb of loop of Henle.
- (3) proximal convoluted tubules.
- (4) all of these

153. Which of the following is present in cortex region?

- (1) Malpighian corpuscles
- (2) Proximal convoluted tubule
- (3) Afferent arteriole
- (4) All of these

154. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Sweat produced by sebaceous glands is a watery fluid containing NaCl, small amount of urea and lactic acid.

Reason R: Sebaceous glands eliminate certain substances like sterols, hydrocarbons and waxes through sebum.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

155. Match List-I with List-II:

List-I		List-II	
(A)	Insects	(I)	Flame cells
(B)	Earthworm	(II)	Antennal glands
(C)	<i>Planaria</i>	(III)	Malpighian tubules
(D)	Crustaceans	(IV)	Nephridia

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-II, B-III, C-I, D-IV
- (4) A-II, B-IV, C-III, D-I

156. The urine formed is a light yellow coloured watery fluid which is **X** having **Y**.

Choose the option which fill the blanks **correctly**.

- | | |
|-----------------------|----------|
| X | Y |
| (1) slightly alkaline | pH-8.0 |
| (2) slightly acidic | pH-2.0 |
| (3) slightly acidic | pH-6.0 |
| (4) slightly alkaline | pH-9.0 |

157. Presence of glucose in urine is called:

- (1) anaemia.
- (2) ketonuria.
- (3) glycosuria.
- (4) asthma.

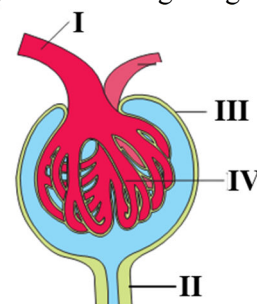
158. Select the **correct** set of organisms that excrete same type of nitrogenous waste.

- (1) Many bony fishes and land snails
- (2) Birds and insects
- (3) Marine fishes and many bony fishes
- (4) Aquatic amphibians and insects

159. Which of the following substances is absorbed by active transport?

- (1) Glucose
- (2) Amino acids
- (3) Nitrogenous wastes
- (4) Both (1) and (2)

160. Identify the structures which is a fine branch of renal artery from the diagram given below.



- (1) I
- (2) II
- (3) III
- (4) IV

161. The **correct** ascending order of percentage proportion of leucocytes in human blood is:

- (1) neutrophils → basophils → lymphocytes.
- (2) monocytes → neutrophils → lymphocytes.
- (3) basophils → lymphocytes → neutrophils.
- (4) monocyte → basophils → eosinophils.

162. Choose the **incorrect** option w.r.t the formation of a blood clot.

- (1) Thrombins are formed from prothrombin by the enzyme complex thrombokinase.
- (2) Fibrins are formed by the conversion of inactive fibrinogens by the enzyme thrombin.
- (3) Coagulum is formed mainly of a network of threads called fibrin.
- (4) Certain factors released by the tissues at the site of injury cannot initiate coagulation.

163. Open circulatory system is present in:

- A. arthropoda.
- B. annelida.
- C. chordata.
- D. mollusca.

Choose the **correct** option:

- (1) B & C only
- (2) A & B only
- (3) B & D only
- (4) A & D only

164. What would be the heart rate of a person if the cardiac output is 6 L and stroke volume is 60 ml?

- (1) 50 beats per minute
- (2) 75 beats per minute
- (3) 100 beats per minute
- (4) 125 beats per minute

165. Given below are two statements.

Statement I: Pericardium is the single layer covering of heart.

Statement II: Fats are absorbed through lymph in the lacteals present in the intestinal villi.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

166. Which of the following options is an **incorrect** match?

	Disorders	Symptoms
(1)	Coronary artery disease	Lumen of arteries becomes broader
(2)	Angina pectoris	Insufficient oxygen to heart
(3)	Cardiac arrest	Heart stops beating
(4)	Heart failure	Heart is not pumping blood effectively enough to meet the needs of the body

167. Which of the following statements is **incorrect**?

- (1) Tricuspid valve guards the opening between the left atrium and the left ventricle.
- (2) Inter-atrial septum separates the right and the left atria.
- (3) The valves in the heart prevents the flow of blood in a backward direction.
- (4) Atrio-ventricular node is located close to the atrio-ventricular septum.

168. The endocrine part of the pancreas is known as:

- (1) acini.
- (2) islets of langerhans.
- (3) hepatocytes.
- (4) adipocytes.

169. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Person with B blood group can receive blood from any blood group.

Reason R: Person with B blood group has anti-A antibodies in their plasma.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

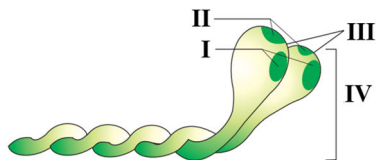
170. Tidal volume and expiratory reserve volume of an athlete is 800 mL and 1000 mL respectively. What will be his expiratory capacity if the residual volume is 1200 mL?

- (1) 1800 mL
- (2) 1700 mL
- (3) 2200 mL
- (4) 3000 mL

171. Which of the following is the smallest contractile unit of skeletal muscle fibre?

- (1) Sarcolemma
- (2) Sarcoplasm
- (3) Sarcomere
- (4) Myosin

172. Refer to the following diagram.



Which of the following represents cross arm?

- (1) I (2) II
(3) III (4) IV
173. In the human body, the type of joint present between the atlas and axis is a:
- (1) hinge joint. (2) ball and socket joint.
(3) pivot joint. (4) gliding joint.

174. Given below are two statements:

Statement I: The hypothalamus is the basal part of diencephalon.

Statement II: The pituitary gland is located in a bony cavity called sella tursica.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

175. Read the following statements (A-D).

- A. The peripheral nervous system consists of the brain and spinal cord.
B. The central nervous system is responsible for processing information.
C. The PNS connects the CNS to limbs and organs through nerves.
D. The brain is divided into three major parts: forebrain, midbrain, and hindbrain.

Choose the most appropriate answer from the options given below:

- (1) Only A, C, and D are correct.
(2) Only A and B are correct.
(3) Only B, C and D are correct.
(4) Only C and D are correct.

176. Which of the following hormone is the main mineralocorticoid in the human body?

- (1) Aldosterone
(2) Cortisol
(3) Adrenaline
(4) Noradrenaline

177. Which of the following is **true** for skeletal muscles?

- (1) They are striated.
(2) They are under voluntary control.
(3) They do not fatigue.
(4) Both (1) and (2)

178. Enlargement of the thyroid gland due to iodine deficiency is called:

- (1) Cretinism (2) Goitre
(3) Myxedema (4) Tetany

179. The gap between two neurons where neurotransmitters are released is called:

- (1) synaptic cleft. (2) nodes of Ranvier.
(3) axon terminal. (4) receptor site.

180. The release of hormones from the posterior pituitary is primarily regulated by:

- (1) direct neural regulation by hypothalamus.
(2) feedback mechanisms from target glands.
(3) hormones secreted by the anterior pituitary.
(4) environmental factors.

181. The outermost layer of the cranial meninges is called:

- (1) arachnoid
(2) pia mater
(3) dura mater
(4) cerebrospinal fluid

182. The myelin sheath around the axon is produced by:

- (1) schwann cells. (2) osteocytes.
(3) chondrocytes. (4) neurons.

183. Read the following statements (A-E).

- A. The anterior pituitary releases growth hormone (GH) in response to growth hormone-releasing hormone.
B. Pelvic girdles meet ventrally to form the pubic symphysis containing fibrous cartilage.
C. The 8th, 9th, and 10th pairs of ribs do not directly attach to the sternum but connect to the seventh rib via hyaline cartilage.
D. A single U-shaped bone called hyoid is present at the base of the buccal cavity.
E. The number of mitochondria are few in white muscle fibers.

How many of the above statements is/are **correct**?

- (1) One
(2) Two
(3) Three
(4) Five

- 184.** Read the following statements (A-D).
- Low GH secretion causes pituitary dwarfism.
 - Ribs have two articulation surfaces at the dorsal end, making them bicephalic.
 - Pineal gland helps to regulate menstrual cycle.
 - Pars intermedia secretes melanocyte stimulating hormone (MSH).
- Choose the most appropriate answer from the options given below:
- Only A, B, and C are correct.
 - Only B and D are correct.
 - Only A, B, and D are correct.
 - A, B, C and D are correct.

- 185.** Match List-I with List-II.

List-I		List-II	
(A)	Amoeboid movement	(I)	<i>Euglena</i>
(B)	Flagellar movement	(II)	Leucocytes
(C)	Myoglobin	(III)	12
(D)	Thoracic vertebrae	(IV)	Red muscle fibers

Choose the **correct** answer from the options given below:

- A-II, B-I, C-IV, D-III
- A-I, B-II, C-III, D-IV
- A-II, B-III, C-IV, D-I
- A-IV, B-III, C-II, D-I

SECTION - B

- 186.** Select the **mismatch** option w.r.t types of neuron.

(1)	Multipolar	-	Cerebral cortex
(2)	Unipolar	-	Embryonic stage
(3)	Bipolar	-	Retina of eye
(4)	Myelinated nerve fibres	-	Somatic neural system

- 187.** How many from the following is/are secreted by anterior pituitary gland?

TSH, GH, PRL, ACTH, GnRH, ADH

- One
- Two
- Four
- Three

- 188.** All of the following statements are correct **except**:

- Impulse transmission across electrical synapses resembles conduction along a single axon.
- Chemical synapses are rare in our system.
- Cerebellum has very convoluted surface in order to provide the additional space for many more neurons.
- Brain stem forms the connections between the brain and spinal cord.

- 189.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The contraction of skeletal muscles is due to the sliding filament mechanism.

Reason R: During skeletal muscle contraction, myosin filaments slide over actin filaments.

In the light of the above statements, choose the **correct** answer from the options given below:

- A is true but R is false.
- A is false but R is true.
- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is not the correct explanation of A.

- 190.** Which of the following disorders is caused by decreased bone mass?

- Osteoporosis
- Arthritis
- Gout
- Muscular dystrophy

- 191.** Which of the following nitrogenous waste requires the largest amount of water for its elimination?

- Urea
- Uric acid
- Ammonia
- Both (1) and (2)

- 192.** Match List-I with List-II.

List-I		List-II	
(A)	Synaptic vesicles	(I)	Hindbrain
(B)	Cerebellum	(II)	Midbrain
(C)	Corpora quadrigemina	(III)	Forebrain
(D)	Corpus callosum	(IV)	Neurotransmitter

Choose the **correct** answer from the options given below:

- A-IV, B-I, C-II, D-III
- A-I, B-II, C-III, D-IV
- A-II, B-I, C-III, D-IV
- A-II, B-III, C-I, D-IV

- 193.** Which of the following is a part of the coxal bone?

- Ilium
- Pubis
- Ischium
- All of these

194. Read the following statements (A-E).
- A. The thyroid gland is composed of follicles and stromal tissues.
 - B. All neurons have a similar basic structure and function.
 - C. Nodes of Ranvier are gaps in the myelin sheath where action potentials are regenerated.
 - D. Basal metabolic rate increases in Grave's disease.
 - E. Excess growth hormone secretion in adults leads to facial disfigurement called acromegaly.

Choose the most appropriate answer from the options given below:

- (1) Only A and B are incorrect.
- (2) Only B and C are incorrect.
- (3) Only C and D are incorrect.
- (4) Only B is incorrect.

195. Which of the following statements about the sympathetic nervous system is **incorrect**?

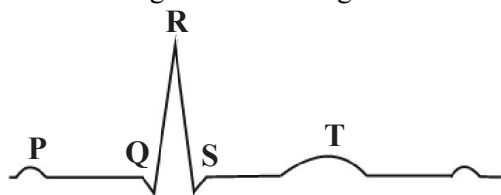
- (1) It increases the heart rate.
- (2) It is a part of the central nervous system.
- (3) It prepares the body for "fight or flight" responses.
- (4) It acts on adrenal gland.

196. Complete the analogy and choose the **correct** option w.r.t vertebral column.

Cranial bones : 8 :: Facial bones: _____

- (1) 14
- (2) 22
- (3) 18
- (4) 7

197. Refer to the given below diagram.



Which of the following represents depolarisation of the atria?

- (1) P
- (2) Q
- (3) T
- (4) R

198. Erythropoietin primarily stimulates the formation of:

- (1) white blood cells.
- (2) platelets.
- (3) red blood cells.
- (4) plasma cells.

199. Which of the following group of hormones includes estradiol and progesterone?

- (1) Peptide hormones
- (2) Steroid hormones
- (3) Iodothyronines
- (4) Amino-acid derivatives

200. Given below are two statements:

Statement I: The resting potential of a neuron is maintained by the sodium-potassium pump.

Statement II: The resting potential is primarily due to the high concentration of K^+ ions inside the neuron.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

